

Metalloids

Also known as semi-metals, the metalloids are an odd collection of elements that show a wide range of chemical and physical properties. Sometimes they act like typical metals, sometimes like non-metals. One example of their behavior as both is their use as semi-conductors in modern electronics.

In the periodic table, the metalloids form a jagged diagonal border between the metals on the left, and the non-metals to the right. Some scientists disagree regarding the exact classification of some elements in this part of the periodic table, precisely because of this in-between status. Some of the elements shown here are toxic, some are more useful than others, some are very common, and some very rare. But they are all solid at room temperature.

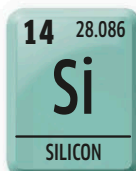
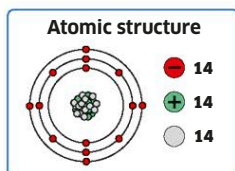


SILICON

Silicium

Discovered: 1823

Most of us are familiar with silicon, even if we don't know it. It is the second most abundant element in the Earth's crust, only after oxygen, and appears in many different silicate minerals. Mixed with other elements, silicon, a typical semi-conductor, is at the heart of the electronics industry—used in microchips and solar panels. Silicone baking molds contain silicon, too.



Silicon sample
Pure silicon, such as this sample refined in a laboratory, shatters easily.



Silicate minerals

Silicon is more or less everywhere, found in the silicate compounds that are better known to us as sand, quartz, talc, and feldspar, and in rocks made up of these minerals. Silicates also include minerals whose crystals make luxurious gems, such as amethyst, opal, lazurite, jade, and emerald. All these contain silica (silicon and oxygen), and sometimes other elements, too (see p.23).



Orthoclase
This feldspar is what gives pink granite its color.

Feldspar minerals

A widespread group of silicate minerals, feldspars contain aluminum as well as silica, and often other elements, too, including calcium, sodium, and potassium. They form common rocks, such as granite. The pretty crystal called moonstone is also a type of feldspar.

Moon mineral
It is not just on Earth that silicates abound. The surface of the moon is made of 45 percent silica.



Genesis rock
Collected on the moon by Apollo 15 in 1971, this rock contains feldspar, a type of silicate mineral.



Silicate sands

Desert sand is chiefly composed of silica, a silicon and oxygen compound with the chemical name silicon dioxide. Sand started out as rock that was gradually broken up and eroded into finer and finer grains. In the Sahara (left), this process started some 7 million years ago.